

WEAKLY-SUPERVISED AUTOMATED AUDIO CAPTIONING VIA TEXT ONLY TRAINING

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ABSTRACT

In recent years, datasets of paired audio and captions have enabled remarkable success in automatically generating descriptions for audio clips, namely Automated Audio Captioning (AAC). However, it is labor-intensive and time-consuming to collect a sufficient number of paired audio and captions. Motivated by the recent advances in Contrastive Language-Audio Pretraining (CLAP), we propose a weakly-supervised approach to train an AAC model assuming only text data and a pre-trained CLAP model, alleviating the need for paired target data. Our approach leverages the similarity between audio and text embeddings in CLAP. During training, we learn to reconstruct the text from the CLAP text embedding, and during inference, we decode using the audio embeddings. To mitigate the modality gap between the audio and text embeddings we employ strategies to bridge the gap during training and inference stages. We evaluate our proposed method on Clotho and AudioCaps datasets demonstrating its ability to achieve a relative performance of up to 83% compared to fully supervised approaches trained with paired target data.¹ Our code is available at: <https://github.com/zelaki/wsac>

Index Terms— Automated audio captioning, multi-modal learning, contrastive learning.

1. INTRODUCTION

Audio-Language tasks have recently gained the attention of the audio community with the introduction of Automated Audio Captioning and Language-Based Audio Retrieval in the DCASE Challenge and the release of publicly available Audio-Language datasets such as Clotho [1] and AudioCaps [2]. The intrinsic relationship between Audio and Language presents an opportunity for the development of models that can effectively establish a shared semantic space for the two modalities. Such an approach has recently achieved great success with models like COALA [3], AudioClip [4], and CLAP [5, 6, 7]. These models use parallel audio-text data to train a joint representation, where the embeddings of audio-text pairs are similar. Such models achieve high accuracy in a zero-shot setting in a variety of tasks including Sound Event Classification, Music tasks, and Speech-related tasks [5].

Automated Audio Captioning (AAC) is a multimodal task that aims to generate textual descriptions for a given audio clip. In order to generate meaningful descriptions, a method needs to capture the sound events present in an audio clip and generate a description in natural language. Training audio captioning models requires

large datasets of audio-caption pairs, and these are challenging to collect. While great effort has been done, the data scarcity issue of audio captioning still withholds. The common datasets in AAC, AudioCaps and Clotho, contain together 50k captions for training, whereas 400k captions are provided in COCO caption [8] for image captioning. Kim et al. [9] observe that due to the limited data, prior arts design decoders with shallow layers that fail to learn generalized language expressivity and are fitted to the small-scaled target dataset. Due to this issue, their performance radically decreases when tested on out-of-domain data. Motivated by these limitations we present an approach to AAC that only requires a pre-trained CLAP model and unpaired captions from a target domain. This alleviates the need for paired audio-text data, and also allows for simple and efficient domain adaptation.

Our approach is inspired by recent advances in zero-shot image captioning [10, 11], that leverage the aligned multi-modal latent space provided by CLIP [12] obviating the need for image data during training and by the recent success of Contrastive Language-Audio models such as CLAP [5] in many downstream tasks. We train a lightweight decoder model to reconstruct texts from their respective CLAP embeddings, and at inference use this decoder to decode the audio embeddings. Our findings align with prior studies in image captioning suggesting that such an approach is suboptimal due to the presence of a phenomenon known as *modality gap* [13].

The *modality gap* suggests that embeddings from different data modalities are located in two completely separate regions of the embedding space of multi-modal contrastive models [13]. To mitigate this issue we employ strategies that have been shown to effectively condense the gap in CLIP embeddings [10, 11] and show that they can be effectively utilized for CLAP models. These strategies can be divided into two categories, strategies that condense the gap during *training* and during *inference*.

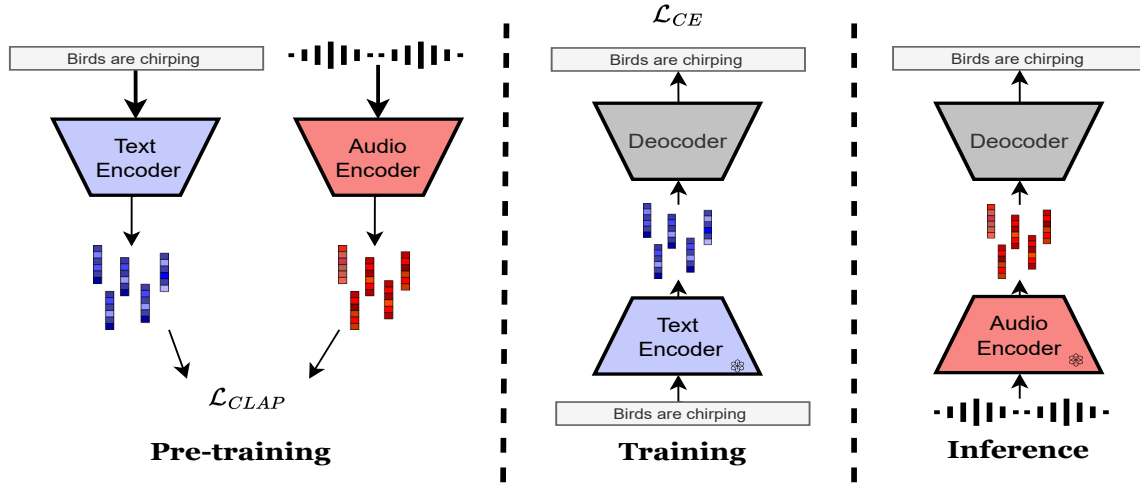
Experiments on Clotho and AudioCaps datasets show that our weakly-supervised approach can achieve comparable performance to prior fully supervised arts, without requiring any target audio data during training. Our contributions can be summarized as follows: (1) We propose **WSAC: Weakly-Supervised Audio Captioning** an AAC approach that requires no auditory in-domain data for training, (2) we demonstrate that the *modality gap* phenomenon is present in CLAP models, and (3) employ methods that effectively mitigate it.

2. TEXT-ONLY TRAINING

Our goal is to learn a model that produces a caption for a given audio clip. Unlike fully supervised approaches, during training we only assume that we have access to a set of target domain captions C . We further assume a pre-trained CLAP model with an audio en-

¹This work was conducted in the framework of the PREMIERE project (No. 101061303) that is funded by the European Union.

Figure 1: Overview of our proposed approach. **Left:** An illustration of the CLAP training paradigm. The encoders are trained to map semantically similar audio-caption pairs to similar embeddings in a joint representation space. **Middle:** Our proposed weakly supervised training. A frozen CLAP text encoder embeds a caption and a decoder learns to reconstruct the caption from its embedding. **Right:** At inference, we decode the audio embedding extracted from a frozen CLAP audio encoder, using the trained decoder.



coder $\mathcal{A}_{clap}$ and a text encoder $\mathcal{T}_{clap}$ trained to project semantically similar audio-text pairs into similar embeddings in a shared embedding space as presented in Fig. 1 (Left). Given an audio clip x_a and text x_t let $\mathbf{z}_a = \mathcal{A}_{clap}(x_a) \in \mathbb{R}^d$ and $\mathbf{z}_t = \mathcal{T}_{clap}(x_t) \in \mathbb{R}^d$ be their embeddings.

First we extract text embeddings $\mathbf{z}_t$ for all $x_t \in \mathcal{C}$, keeping $\mathcal{T}_{clap}$ frozen. During training, our goal is to learn a network that inverts the CLAP text encoder $\mathcal{T}_{clap}$. We use a textual decoder D consisting of a mapping network f and an auto-regressive language model, to reconstruct the original text x_t from the CLAP text embedding $\mathbf{z}_t$. Following recent work [9], we train our decoder using the prefix language modeling paradigm. Specifically, after passing the text embedding through the mapping network f we regard $\mathbf{p} = f(\mathbf{z}_t)$ as a prefix to the caption. Given a text $t = \{w_1, w_2, \dots, w_T\}$, our objective is to minimize the autoregressive cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^T \log D(w_i | w_{<i}, \mathbf{p}) \quad (1)$$

Since the CLAP text embedding is optimized to be similar to the CLAP audio embedding, we can directly infer the text decoder using the audio embeddings $\mathbf{z}_a$ without any pairwise training on the target dataset. The training and inference stages are presented in Fig. 1 (middle) and (right) respectively.

3. STRATEGIES TO BRIDGE THE MODALITY GAP

Directly employing the audio embeddings to infer D is not optimal due to the presence of the modality gap. Fig. 2 is a visualization of generated embeddings from the pre-trained CLAP model from the Clotho training set. Paired inputs are fed into the pre-trained model and the embeddings are visualized in 2D using T-SNE [14]. This visualization clearly demonstrates the presence of the modality gap phenomenon, as a noticeable gap separates the paired audio and text embeddings. To address this issue, we utilize strategies that have demonstrated success in bridging the modality gap in CLIP

embedding space [10, 11, 13]. We show that these strategies can be adopted for CLAP and show their effectiveness in mitigating the modality gap. These approaches can be divided into two categories: Bridging the gap either during the training phase or during the inference phase.

3.1. Training strategies

Attempting to reduce the modality gap during training we adopt the following strategies: (a) Noise injection [10], and Embedding Shift [13]. These strategies aim to narrow the disparity between the modality used to train the decoder, which is text, and the target modality, which is audio.

3.1.1. Noise injection

In [10], the authors show that injecting the text embedding with Gaussian noise during training has the effect of creating a region in the embedding space that will map to the same caption. This method assumes that the corresponding audio embedding is more likely to be inside this region. Following [10], we add zero-mean Gaussian noise of standard deviation σ to the text embedding before feeding it to the decoder. We set σ to the mean L_{inf} norm of embedding differences between five captions that correspond to the same audio. Since we assume no access to target audio data we estimate σ using 50 audio-caption pairs from the WavCaps dataset [7]. Thus the prefix in Eq. 1 becomes $\mathbf{p} = f(\mathbf{z}_t + \mathbf{n})$, where $\mathbf{n} \in \mathbb{R}^d$ is a random standard Gaussian noise with standard deviation σ .

3.1.2. Embedding shift

Building upon the findings of [13], who investigated the impact of shifting embeddings in various multi-modal contrastive learning models on downstream tasks, we propose a method to align the text embeddings with the audio embeddings during training. First, we define the modality gap following [13], as the difference between the center of audio embeddings and text embeddings:

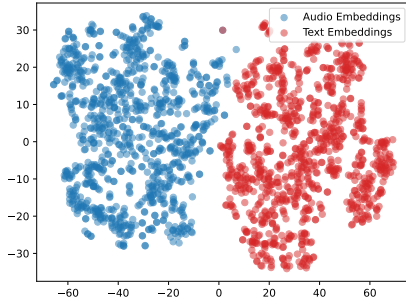


Figure 2: Visualization of audio and text embedding pairs randomly sampled from the Clotho training set. The modality gap phenomenon is present as the audio and text modalities are embedded in two completely separate regions.

$$\Delta_{\text{gap}} = \frac{1}{n} \sum_{i=1}^n \mathbf{z}_{a_i} - \frac{1}{n} \sum_{i=1}^n \mathbf{z}_{t_i} \quad (2)$$

Then, we shift every text embedding toward closing the modality gap, and thus the prefix in Eq. 1 becomes $\mathbf{p} = f(\mathbf{z}_t + \Delta_{\text{gap}})$.

3.2. Inference strategies

At inference, we adopt two training-free strategies proposed in [11], and map an audio embedding extracted from the CLAP audio encoder $\mathcal{A}_{\text{clap}}$ into the text embedding space. For both strategies, we will assume a decoder D trained on some target data as described in Section 2 and a set of text embeddings obtained from the target training set that we will refer to as *Memory*, $\mathcal{M} = \{\mathbf{z}_t^1, \mathbf{z}_t^2, \dots, \mathbf{z}_t^N\}$, where N is the size of the training set.

3.2.1. Nearest-neighbor decoding

A straightforward strategy that can be adopted at inference time to mitigate the modality gap is to use the nearest text embedding as the prefix, instead of the audio embedding. We calculate the cosine similarity between the audio embedding $\mathbf{z}_a$ and the text embeddings in $\mathcal{M}$ and decode with the most similar:

$$\mathbf{p} = \mathbf{z}_i \mid i = \underset{\mathbf{z}_t \in \mathcal{M}}{\operatorname{argmax}} \operatorname{sim}(\mathbf{z}_a, \mathbf{z}_t) \quad (3)$$

Where $\operatorname{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|}$. Since the decoder is trained to reconstruct the original text conditioned on the text embedding, nearest-neighbor decoding can be successful if a sufficiently similar text embedding is present in $\mathcal{M}$.

3.2.2. Projection-based decoding

A better approach is to project the audio embedding into the text embedding space. This involves obtaining the representation of the audio embedding, by combining the embeddings in $\mathcal{M}$ through a weighted combination.

$$\mathbf{p} = \sum_{i=1}^{|\mathcal{M}|} w_i * \mathbf{z}_{t_i} \quad (4)$$

The weights w_i for these text embeddings are determined by calculating the cosine similarity between the audio embedding $\mathbf{z}_a$ and each embedding in $\mathcal{M}$. Following [11] the similarity is then scaled by a temperature parameter τ and normalized using a softmax function:

$$w_i = \frac{\exp(\operatorname{sim}(\mathbf{z}_a, \mathbf{z}_{t_i})/\tau)}{\sum_{j=1}^{|\mathcal{M}|} \exp(\operatorname{sim}(\mathbf{z}_a, \mathbf{z}_{t_j})/\tau)} \quad (5)$$

4. EXPERIMENTS

4.1. Data

We conduct experiments using two benchmarks, AudioCaps and Clotho. AudioCaps contains 50k, 10-second audio clips sourced from Audioset [15]. Each audio is annotated with one caption in the training set and five captions in the evaluation set. Clotho consists of 4981 audio samples of 15 to 30 seconds duration. Each audio is annotated with five captions. We follow the standard recipes of training, validation, and test splits on each dataset for our experiments. To adhere to a weakly-supervised setting we assume no access to audio data in the training and validation sets.

4.2. Experimental setup

To extract audio and text embeddings we employ a frozen CLAP model² trained on WavCaps [7]. The audio encoder is a CNN14 from Pre-trained Audio Neural Networks (PANNs) [16], and the text encoder is a BERT-based model [17]. We choose this model as the embedding extractor because AudioCaps and Clotho datasets were not included in its training set. This choice is made under the assumption that target audio data are unavailable for training purposes. The decoder D consists of a mapping network f which is a 2-layered MLP, and the language model which is a 4-layer Transformer [18] with 4 attention heads. The size of the hidden state is 768. The decoder D is trained from scratch on the target captions. The noise variance for *Noise Injection* training is set to $\sigma^2 = 0.013$. We train the proposed model for 30 epochs using Adam optimizer [19] and a batch size of 64. The learning rate is linearly increased to 2×10^{-5} in the first five epochs using warm-up, which is then multiplied by 0.2 every 10 epochs. We use greedy search for decoding.

4.3. Compared methods and evaluation metrics

Since no previous work has addressed AAC in similar supervision settings we compare our methods against fully supervised approaches trained on paired data. Koh et al. [23] use a latent space similarity objective and train a model with a PANNs encoder and a transformer decoder. Xu et al. [22] design a GRU for the decoder. Mei et al. [20] propose a full transformer encoder-decoder architecture. Gontier et al. [21] utilize a pre-trained language model based on BART [21], and finetune it for AAC using guidance from Audioset tags. Kim et al. [9] propose prefix tuning for AAC learning a prefix to guide the caption generation of a frozen GPT-2 [24]. Mei et al. [7] utilize a CLAP audio encoder pre-trained on WavCaps and a BART decoder achieving state-of-the-art results in both Clotho and AudioCaps. All the methods in this work are evaluated by the metrics widely used in the captioning tasks, including BLEU [25], METEOR [26], ROUGE-L [27], CIDEr [28], SPICE [29], and SPIDEr [30].

²<https://github.com/XinhaoMei/WavCaps/tree/master>

Table 1: **Results on AudioCaps and Clotho.** We report results for fully supervised methods trained on audio-caption pairs, and our proposed methods trained only on captions. WSAC is our baseline approach presented in Section 2. We refer to *Noise injection* as NI, *Embedding shift* as ES, *Nearest-neighborhood decoding* as NND and, *Projection-based decoding* as PD. We highlight the best results for fully and weakly supervised methods with underline and **bold** respectively.

Dataset	Supervision	Method	BLEU ₁	BLEU ₂	BLEU ₃	BLEU ₄	METEOR	ROUGE _L	CIDEr	SPICE	SPIDEr
Audiocaps	Audio-Caption Pairs	Mei et al. [20]	0.647	0.488	0.356	0.252	0.222	0.468	0.679	0.160	0.420
		Kim et al. [9]	0.713	0.552	0.421	0.309	0.240	0.503	0.733	0.177	0.455
		Gontier et al. [21]	0.699	0.523	0.380	0.266	0.241	0.493	0.753	0.176	0.465
		Mei et al. [7]	<u>0.707</u>	-	-	<u>0.283</u>	<u>0.250</u>	<u>0.507</u>	<u>0.787</u>	<u>0.182</u>	<u>0.485</u>
	Captions Only	WSAC	0.574	0.398	0.267	0.167	0.222	0.426	0.493	0.155	0.324
		WSAC+NI	0.662	0.477	0.328	0.216	0.223	0.46	0.579	0.155	0.367
		WSAC+ES	0.653	0.458	0.300	0.185	0.214	0.451	0.540	0.154	0.347
		WSAC+NND	0.643	0.457	0.312	0.198	0.231	0.454	0.548	0.166	0.357
		WSAC+PD	0.698	0.511	0.357	0.232	0.241	0.479	0.633	0.173	0.403
Clotho	Audio-Caption Pairs	Xu et al. [22]	0.556	0.363	0.242	0.159	0.169	0.368	0.377	0.115	0.246
		Koh et al. [23]	0.551	0.369	0.252	0.168	0.165	0.373	0.380	0.111	0.246
		Kim et al. [9]	0.560	0.376	0.253	0.160	0.170	0.378	0.392	0.118	0.255
		Mei et al. [7]	<u>0.601</u>	-	-	<u>0.180</u>	<u>0.185</u>	<u>0.400</u>	<u>0.488</u>	<u>0.133</u>	<u>0.310</u>
	Captions Only	WSAC	0.462	0.282	0.173	0.102	0.166	0.343	0.265	0.113	0.189
		WSAC+NI	0.525	0.314	0.193	0.118	0.164	0.352	0.315	0.113	0.214
		WSAC+ES	0.546	0.332	0.203	0.120	0.159	0.353	0.301	0.109	0.205
		WSAC+NND	0.498	0.294	0.179	0.106	0.166	0.338	0.332	0.113	0.222
		WSAC+PD	0.532	0.324	0.200	0.118	0.174	0.354	0.371	0.123	0.247

4.4. Results and Discussion

In this section, we present the results of our proposed methods on the performance metrics and compare them with fully supervised arts. Additionally, we illustrate the effectiveness of each strategy in reducing the modality gap. As shown in Table 1 our methods demonstrate comparable performance to prior state-of-the-art models despite never encountering in-domain audio data during training. We present the results of our baseline approach described in Section 2 and the results of the baseline approach in conjunction with the strategies presented in Section 3. It is evident that all the strategies boost the performance of our baseline approach in both evaluation sets. Interestingly the *inference strategies* outperform the *training strategies* in most cases. We hypothesize that this is because they utilize the *Memory M* which consists of in-domain text embeddings in order to bridge the modality gap. Our best-performing method, namely *Projection-based decoding* achieves 80% and 83% of the SPIDEr performance of the current fully supervised state-of-the model in Clotho and AudioCaps evaluation sets respectively. Additionally *Projection-based decoding* matches the performance of the of fully-supervised approaches proposed by Kim et al. [9]. Koh et al. [23] and Xu et al. [22] in the Clotho evaluation set.

Visualization of embeddings: To further examine the effectiveness of the proposed strategies we illustrate the embeddings in 2D space using t-SNE in Fig. 3. In Fig. 3a and 3b we randomly sample audio and text embeddings from the Clotho training set after applying *Noise Injection* and *Embedding Shift* to the text embeddings. Fig. 3c and 3d illustrate randomly selected text embeddings from the Clotho evaluation set, alongside the embeddings utilized for decoding, namely the nearest neighbors and the projections, rather than the paired audio embeddings. It is evident that all strategies are effective in condensing the modality gap showcased in Fig. 2, where the audio and text modalities are embedded at arm’s length in their shared representation space.

5. CONCLUSION AND FEATURE WORK

In this work, we propose a weakly-supervised approach for Automated Audio Captioning that requires a pre-trained CLAP model and only additional text data to train on a target domain. Our method

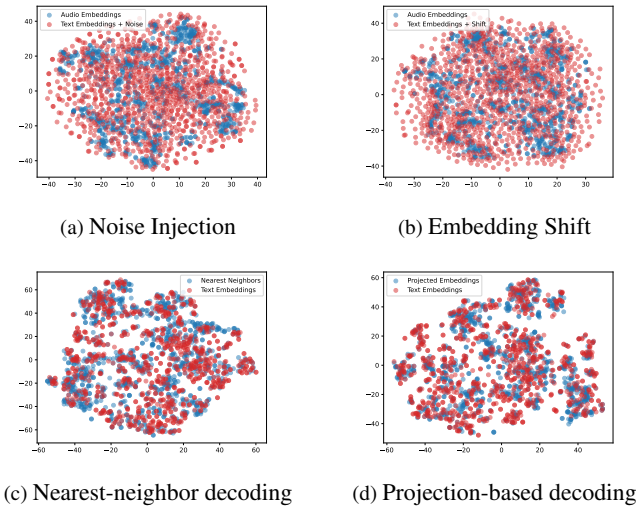


Figure 3: TSN-E visualizations of the embedding space after applying the strategies presented in Section 3.

alleviates the necessity of paired data in a target domain, which are hard to collect. We demonstrate that by leveraging the shared embedding space of CLAP we can learn to reconstruct the text from the CLAP text embedding and during inference decode using the audio embeddings. We show that such an approach is suboptimal due to the presence of a modality gap and adopt strategies that effectively mitigate it. Our best-performing method achieves comparable results to prior arts trained in a fully supervised manner. For future work, we plan to study the effectiveness of our proposed approach on other tasks, such as Music Captioning and Audio Question Answering. We further aim to train a mapping network to learn the gap between the two modalities in a supervised manner.

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